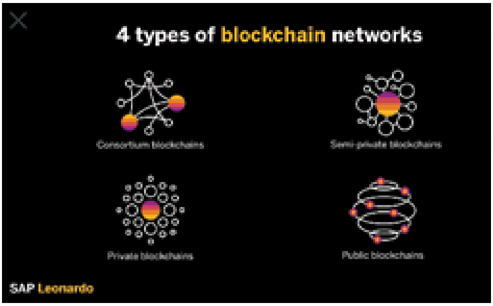

as it's always unique. Every hash identifies its block and its content through this hash or fingerprint.

As soon as a block is created, its hash is calculated. Whenever there is some or other change in data inside the block, the hash will automatically be disturbed. In other words, the hash is a vital block element, especially when you want to detect the changes to blocks. If you find a change in fingerprint or hash, it's no longer the same block.

Now the third and the last element of the block is the hash of the previous block. Let us first tell you that this technique of blockchain is why Satoshi Nakamoto was able to trust this system for bitcoin as it ensures the utmost security of data.

In a blockchain, the hash of a previous block effectively creates a chain of blocks in a tamper-proof sequence as the hash is exceptionally sensitive. Each of these blocks stores the block's fingerprint before it, along with its own unique fingerprint. Also, any information you put into it will be stored in these blocks.

The hash interconnects these blocks, so if you try to tamper with the data of any block, the fingerprint or the hash of that particular block will automatically change, leading to the change in all the following blocks as these are interconnected, as we mentioned earlier. Eventually, the entire blockchain will be destroyed. Hence, it's nearly impossible to tamper with the data in the block as once the data is defined, it cannot be altered later.

But that's not enough! Computers these days are exceptionally smart to calculate hundreds and thousands of hashes per second. With the right skills, you can effectively tamper with a block and recalculate all the hashes of the following blocks to make your blockchain valid again.

To eliminate this possibility, blockchain offers something that is known as proof-of-work, which is an algorithm that slows down the creation of new blocks. These blocks are managed by the players who execute proof-of-work, popularly known as "miners." Every new block is accepted by the network that takes about 10 minutes for the complete process.